04/01 Multivariate Analysis

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Books - 1) Anderson
2) Johnson K ..... → Application.

30, 30, 40 [Quiz, Mid sem, End sem]

What is Multivariate Analysis ?
Does outcome depends on one variable ? (not)

$\sum_{i=1}^{n} x_i^2$ [scalar]

$\underline{x} = (x_1, \dots, x_n)$

$\underline{x}\ \underline{x}'$, $\quad x'Ax \geq 0$

$$\text{health} = \begin{pmatrix} Ht \\ wt \\ BP \\ BS \\ \vdots \end{pmatrix}$$

$X \to \mu X,\ \sigma_X^2 \qquad \delta_{XY}$

$Y \to \mu Y,\ \sigma_Y^2$

C_{YZ}

$Z \to \mu_Z,\ \sigma_Z^2$

C_{XZ}

$$\begin{array}{c|ccc} & X & Y & Z \\ \hline X & \sigma_X^2 & \sigma_{XY}^2 & \sigma_{XZ}^2 \\ Y & & \sigma_Y^2 & \sigma_{YZ}^2 \\ Z & & & \sigma_Z^2 \end{array}$$

06/01

Multivariate Normal distribution

(difference b/w) Univariate / Multivariate (more than 1 r.v.)

Univariate	Multivariate
X	$X\ Y\ Z \longrightarrow \underline{X} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$
X : height	height, weight, age
$X_1 = 30 \quad X_2 = 40$	$\underline{X_1} = \begin{pmatrix} 30 \\ 50 \\ 25 \end{pmatrix} \quad \underline{X_2} = \begin{pmatrix} 40 \\ 55 \\ 30 \end{pmatrix}$
$X_1, \dots, X_n$: Random variable	$\underline{X_1}, \dots \underline{X_n}$ (now this will be vectors)
$X \sim N(\mu, \sigma^2)$	

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$ (Legendra gave Normal dist^n)

(to convert it into multivariate)

$X \quad \mu \quad \sigma^2 \longrightarrow \underset{\text{x data vector}}{\underline{X}} \quad \underset{\text{mean vector}}{\underline{\mu}} \quad \underset{\text{var-cov.}}{\Sigma}$

$$\underbrace{f_X(x)}_{\text{univariate}} = \underbrace{\frac{1}{\sigma\sqrt{2\pi}}}_{\text{constant}} \exp\left[-\frac{1}{2}\underbrace{(x-\mu)(\sigma^2)^{-1}(x-\mu)}_{\text{Quadratic form}}\right]$$

quadratic forms - scalar

$\sum x_i^2 = x'Ax$

$x'x$ or xx'

see order, check whether scalar or vector.

$\underline{X} \longrightarrow p$ variables $\quad X_1, \ldots, X_p \quad \underline{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_p \end{pmatrix}$

$\underline{x}_1, \ldots, \underline{x}_n \qquad \begin{matrix} x_{11} \\ x_{12} \\ \vdots \end{matrix}$

μ be replaced by vector b

(M.V.N.D with complex r.v.)
Electrical department

$\mu \longrightarrow \underline{b}, \qquad \sigma^2 \longrightarrow A$

$$\underline{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_p \end{pmatrix} \qquad A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ \vdots & & & \vdots \\ a_{p1} & \cdots & \cdots & a_{pp} \end{pmatrix}$$

A = P.D (positive definite) matrix
A = full rank

$$(x-\mu)(\sigma^2)^{-1}(x-\mu) = (\underline{x}-\underline{b})' A (\underline{x}-b)$$
$$= \Sigma \Sigma \, a_{ij} \, (x_i - b_i)(x_j - b_j)$$

$$f_{X_1, \ldots, X_p}(x_1, \ldots, x_p) = f_X(x)$$
$$= \underset{\substack{\downarrow \\ \text{constant}}}{K} \exp\left[-\frac{1}{2}(\underline{x}-\underline{b})' A (\underline{x}-b)\right]$$

$K > 0$, integral over entire p-dimensional eucledean space of $x_1, \ldots, x_p$ is unity.

$f(x_1, \ldots, x_p) \geqslant 0$ (non-negative)
Bounded, $\quad f(x_1, \ldots, x_p) \leq k$

~~Fix~~ Find k,

$$\underbrace{K \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}(\underline{x}-b)' A (x-b)\right]}_{K^*} = 1$$

$$K K^* = 1$$

$\because$ A is p.d matrix, so $\exists$ a non-singular matrix c s.t. $\quad c' A c = I$

$$c^{-1\prime} c' A c c^{-1} = c'^{-1} c^{-1}$$

$$A = (cc')^{-1} \quad \text{or} \quad A^{-1} = cc'$$

A is p.d., replica of σ^2.

$\Sigma^{1/2} \qquad |\Sigma^{1/2}| \longrightarrow$ scalar

$f(y)\,dy = f(x)\,dx$

$f(y) = f(x) \dfrac{dx}{dy}$

$\underline{x} - \underline{b} = c\underline{y}$ $\qquad \underline{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_p \end{pmatrix}$

$(\underline{x}-\underline{b})' A (\underline{x}-b) = \underline{y}' c' A c \underline{y} \qquad = y'y$

<u>Jacobian of transformation</u> $J = \text{mod } |c|$ $\hookrightarrow$ determinant

$$|J| = \frac{\partial(x_1, \dots, x_p)}{\partial(y_1, \dots, y_p)} = |c|$$

$$f_{\underline{Y}}(\underline{y}) = f_{\underline{X}}(\underline{x}) \; |J|$$

$$f_{\underline{Y}}(\underline{y}) = K . \exp\left[-\frac{1}{2} y' c' A c y\right] |c| \longrightarrow \text{scalar}$$

$$= K \, |cc'|^{1/2} \exp\left[-\frac{1}{2} y' y\right]$$

$$= K \, |A^{-1}|^{1/2} \exp\left[-\frac{1}{2} y' y\right]$$

$$\int_{\mathbb{R}^p} f_{\underline{Y}}(\underline{y}) \, d\underline{y} = 1$$

$$K \, |A^{-1}|^{1/2} \int_{\mathbb{R}^p} \exp\left[-\frac{1}{2} \sum_{i=1}^{p} y_i^2\right] \prod_{i=1}^{p} dy_i = 1$$

$$K \, |A^{-1}|^{1/2} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \prod \exp\left(-\frac{1}{2} y_i^2\right) dy_i = 1$$

$$Y_i \sim N(0,1)$$

$$\int \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2} y_i^2\right] = 1$$

$$\int \exp\left[-\frac{1}{2} y_i^2\right] = \sqrt{2\pi}$$

$$K \, |A^{-1}|^{1/2} \prod_{i=1}^{p} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2} y_i^2\right) dy_i = 1$$

$$K \, |A^{-1}|^{1/2} \left(\sqrt{2\pi}\right)^p = 1$$

$$K = \frac{1}{|A^{-1}|^{1/2} (2\pi)^{p/2}}$$

$$E(\underline{y}) = \begin{pmatrix} E(y_1) \\ \vdots \\ E(y_p) \end{pmatrix}$$

$\underline{b}$? $\qquad \underline{y} = (y_1 \; \dots \; y_p)'$

$$E(Y_i) = \int\int \dots \int y_i \prod_{j=1}^{p} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} y_j^2\right) dy_1 \dots dy_p$$

$$= \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y_i \exp\left(-\frac{1}{2} y_i^2\right) dy_i}_{\text{odd fn } = 0} \cdot \underbrace{\left(\prod_{j=1, \; j\neq i}^{p} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} y_j^2\right) dy_j\right)}_{=1}$$

$E(Y_i) = 0$

$c. E(\underline{y}) = 0 \quad = E(\underline{x} - \underline{b})$

$\Rightarrow \underline{b} = E(\underline{x})$ (mean vector)

$$= \begin{pmatrix} E(x_1) \\ \vdots \\ E(x_p) \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix} = \mu$$

$A = ?$

$$E(x-b)'(x-b) = c.E(yy').c'$$

$$= c.E \begin{pmatrix} y_1^2 & y_1 y_2 & \dots & y_1 y_p \\ \vdots & & & \\ y_p y_1 & \dots & \dots & y_p^2 \end{pmatrix} . c'$$

diagonal and non-diag. element, if we find both exp. separately then put together

$E(Y_i^2) \qquad E(Y_i Y_j)$

$$E(Y_i Y_j) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} Y_i Y_j . \exp\left(-\frac{1}{2} y_i^2\right) . \exp\left(-\frac{1}{2} y_j^2\right) . \prod_{\substack{k\neq i \\ k \neq j}} \exp\left(-\frac{1}{2} y_k\right)^2 \underline{dy}$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} Y_i \exp\left(-\frac{1}{2} Y_i^2\right) dy_i \int_{-\infty}^{\infty} Y_j \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} Y_j^2\right) dy_j$$

$$\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{\substack{k= \\ k\neq i \\ k\neq j}}^{p} \dots$$

$= 0$ (odd fn.)

$$E(Y_i^2) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} Y_i^2 \exp\left(-\frac{1}{2} Y_i^2\right) dy_i$$

$$\prod_{\substack{k=1 \\ k \neq i}}^{p} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} Y_k^2\right) dy_k$$

$$= 1$$

$$E(\underline{x}-\underline{b})'(\underline{x}-b) = c\, E(yy')\, c'$$

$$= c \begin{pmatrix} 1 & \cdot & \cdots & \cdot & 0 \\ \vdots & \cdot & \vdots & \vdots & \\ 0 & \cdot & \cdot & \cdot & 1 \end{pmatrix} c'$$

$$= cc' \quad = A^{-1} \quad = \Sigma$$ (covariance matrix of b)

$$E(\underline{x}-\underline{\mu})(x-\mu)' = \Sigma$$

$$f_{\underline{x}}(\underline{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left[-\frac{1}{2} (\underline{x}-\underline{\mu}) \Sigma^{-1} (\underline{x}-\underline{\mu})\right]$$

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$

Σ = Not full rank, singular distribution.

<u>10/01/24</u>

<u>Theorem.</u> If the density of a p-dim (random vector / variable) r.v. $\underline{X}$ is

$$\frac{\sqrt{|A|}}{(2\pi)^{p/2}} \exp\left\{-\frac{1}{2}(\underline{x}-\underline{b})' A (\underline{x}-\underline{b})\right\}$$

then $E(\underline{X}) = b$ and $\text{cov}(\underline{X}) = A^{-1}$.

Conversely, given a $\underline{\mu}$ and Σ (positive definite vector) > 0, there is a MND s.t. $E(\underline{X}) = \mu$,

$$\text{Cov}(\underline{X}) = \Sigma = \frac{|\Sigma|^{-1/2}}{(2\pi)^{p/2}} \exp\left\{-\frac{1}{2}(\underline{x}-\underline{\mu})' \Sigma^{-1} (\underline{x}-\underline{\mu})\right\}$$

X: is said to ~~be~~ have a MND if its pdf is

$\theta = (\theta_1, \ldots, \theta_K)'$, $f(\theta)$: real valued fn. of $\theta_1, \ldots, \theta_K$

$$\frac{\partial f(\theta)}{\partial \theta} = \begin{bmatrix} \frac{\partial}{\partial \theta_1} f(\theta) \\ \vdots \\ \frac{\partial f(\theta)}{\partial \theta_k} \end{bmatrix}$$

If A: $k \times k$ symmetric matrix $f(\theta) = \theta' A \theta$

$$\frac{\partial f(\theta)}{\partial \theta} = (A + A')\,\theta = 2A\theta$$

A is symmetric matrix.

Characterstic function

Univariate : $\phi(\theta) = E[\exp(i\theta x)] = \int e^{i\theta x} f(x)\, dx$

Multivariate : $\phi(\underline{\theta}) = E[\exp(i\underline{\theta}'\underline{x})]$ (need scalar)

$$= \int e^{i\underline{\theta}'\underline{x}} f_x(\underline{x})\, d\underline{x}$$

Properties in multivariate : i) $\phi(0) = 1$ (vector quantity)

ii) $|\phi(\theta)| \leq 1$

Characteristic fn. always exist but MGF may or may n

Theorem : Uniqueness Theorem

Two r.v. having same characteristic fn iff they have the same dist^n.

Inversion theorem

Given a characteristic fn. we can find the pdf by inversion thm. (Diff the ch fn., put $\theta = 0$)

Ch. fn. of MND

$$\underline{X} \sim MVN(\mu, \Sigma)$$

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$

$$\phi(\underline{t}) = E[\exp(i\underline{t}'\underline{x})] = \int \dots \int e^{i\underline{t}'\underline{x}} \cdot \frac{1}{2\pi} \dots$$

Let $\underline{x} - \underline{\mu} = c\,\underline{y}$

$\phi(t) = E[\exp\, i\underline{t}'(c\underline{y} + \underline{\mu})] = \exp(i\underline{t}'\underline{\mu})\; E[\exp(i\underline{t}'c\underline{y})]$

vector

$u' = t'c$

$= \exp(i\underline{t}'\underline{\mu})\; E[\exp(i\underline{u}'\underline{y})]$

assumption that it is a column vector

$\sum_{j=1}^{p} u_j y_j$ ← sum of $u_i \times$ iid normal

we have information about y_i but we have y over here.

$$= \exp(i\underline{t}'\underline{\mu}) \prod_{j=1}^{p} E[\exp(iu_j y_j)]$$

$X \sim N(\mu, \sigma^2)$ $\quad \phi_x(t) = \exp(i\mu t - \frac{1}{2}\sigma^2 t^2)$

$\mu = 0, \; \sigma^2 = 1$ $\quad \phi_x(t) = \exp(-\frac{1}{2}t^2)$

Treat $u_j = t, \; y_j = z$

$$E[\exp(iu_j y_j)] = \exp\left(-\frac{1}{2}u_j^2\right)$$

$$\phi(t) = \exp(i\underline{t}'\underline{\mu}) \prod_{j=1}^{p} \exp\left(-\frac{1}{2}u_j^2\right)$$

$\Sigma u_j^2 = u^T u$

$$= \exp(i\underline{t}'\underline{\mu})\; \exp\left(-\frac{1}{2}\underline{u}'\underline{u}\right)$$

$$= \exp\left(i\underline{t}'\underline{\mu} - \frac{1}{2}\underline{u}'\underline{u}\right)$$

$u'u = t'cc't$
$\quad = t'\Sigma t$

$$= \exp\left(it'\underline{\mu} - \frac{1}{2}t'\Sigma t\right)$$

We have to switch between univariate and multivariate. There is no uniform rule, but when there is only a dire need then only switch in between.

$$f_{\underline{x}}(\underline{\mu}, \Sigma) = \underbrace{\frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}}} \exp\left[-\frac{1}{2}(Q_1 + Q_2)\right]$$

$$= \frac{1}{(2\pi)^{q/2} |\Sigma_{11}|^{1/2}} \exp\left(-\frac{Q_1}{2}\right) \frac{1}{(2\pi)^{\frac{p-q}{2}} |\Sigma_{22}|^{1/2}} \exp\left(-\frac{Q_2}{2}\right)$$

$$= N_q(\underline{\mu_1}, \Sigma_{11}) . \quad N_{p-q}(\underline{\mu_2}, \Sigma_{22})$$

$\underline{X}_1 \sim N_q(\underline{\mu_1}, \Sigma_{11}) \qquad \underline{X}_2 \sim N_{p-q}(\underline{\mu_2}, \Sigma_{22})$

$f_x(x) = \int_{-\infty}^{\infty} f_{xy}(xy)\, dy$

<u>Alternate</u>

$$f(\underline{X}_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} N_p(\underline{x}, \mu, \Sigma) \; dx_{q+1} \cdots \cdots dx_p$$

$$= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \boxed{N_p(X_1, \mu_1, \Sigma_{11})} \; \underbrace{N_{p-q}(X_2, \mu_2, \Sigma_{22})}_{=1} . \; dx_{q+1} \cdots dx_p$$

$$= N_p(X, \mu, \Sigma)$$

<u>Case</u> ii) $\underline{X}_1$ and $\underline{X}_2$ are not independent $\Sigma_{21} \neq 0$.

why did we use $X - \mu = cy$?
X has covariance matrix $\neq 0$, Y's were independent
The transformed variables becomes independent.

$\underline{Y}_1 = \underline{X}_1 + B X_2$ (value of B s.t. Y_1 and Y_2 are ind.)

$\underline{Y}_2 = \underline{X}_2$

choose B s.t. $\underline{Y}_1$ and $\underline{Y}_2$ are univariate.
(if two variables are uncorrelated then they are ind.)

$E\,(Y_1 - E(Y_1))(Y_2 - E(Y_2))' = 0$

this is going to be a cov-matrix.

$E\left[(\underline{X}_1 + B\underline{X}_2) - E(\underline{X}_1 + B\underline{X}_2)\right]\left[\underline{X}_2 - E(\underline{X}_2)\right]' = 0$

$E\left[(\underline{X}_1 - E(\underline{X}_1)) + B(\underline{X}_2 - E(\underline{X}_2))\right]\left[\underline{X}_2 - E(\underline{X}_2)\right]' = 0$

$\Sigma_{12} + B\,\Sigma_{22} = 0 \quad \Rightarrow \quad \left\{ B = -\frac{\Sigma_{12}}{\Sigma_{22}} \right\}$ error

$$\Sigma_{12} + B\,\Sigma_{22} = 0 \quad \Rightarrow \quad B\,\Sigma_{22} = -\Sigma_{12}$$

$$\Rightarrow B\,\Sigma_{22}\,\Sigma_{22}^{-1} = -\Sigma_{12}\,\Sigma_{22}^{-1} \quad \Rightarrow \quad B = -\Sigma_{12}(\Sigma_{22})^{-1}$$

$$\underline{Y_1} = \underline{X_1} - [\Sigma_{12}(\Sigma_{22})^{-1}\,\underline{X_2}]$$

$$\underline{Y_2} = \underline{X_2} \qquad Y_1 \sim N(E(Y_1),\ Cov(Y_1))$$

$$\underline{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} I & -\Sigma_{12}(\Sigma_{22})^{-1} \\ 0 & I \end{pmatrix} \underline{X}$$

Non-singular transformation ? (Yes)

[if it is a non-singular matrix it follow MVN]

Therefore $\underline{Y} \sim N(\quad)$

$$E\begin{pmatrix} \underline{Y_1} \\ \underline{Y_2} \end{pmatrix} = \begin{pmatrix} I & -\Sigma_{12}\,\Sigma_{22}^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} \underline{\mu_1} \\ \underline{\mu_2} \end{pmatrix}$$

$$= \begin{pmatrix} \underline{\mu_1} - \Sigma_{12}\,\Sigma_{22}^{-1}\,\underline{\mu_2} \\ \underline{\mu_2} \end{pmatrix} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

$$Cov(Y) = E\begin{bmatrix} (\underline{Y_1} - \underline{V_1})(\underline{Y_1} - \underline{V_1})' & (\underline{Y_1} - \underline{V_1})(\underline{Y_2} - \underline{V_2})' \\ (\underline{Y_2} - \underline{V_2})(\underline{Y_1} - \underline{V_1})' & (\underline{Y_2} - \underline{V_2})(\underline{Y_2} - \underline{V_2})' \end{bmatrix}$$

$$= \begin{pmatrix} \Sigma_{11} - \Sigma_{12}\,\Sigma_{22}^{-1}\,\Sigma_{21} & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

$$\Sigma_{22\cdot1} = \Sigma_{22} - \Sigma_{21}\,\Sigma_{11}^{-1}\,\Sigma_{12}$$

$Cov(Y)$ has only diagonal matrix, off-diagonal is 0.

$$\underline{Y_1} \sim N(V_1,\ \Sigma_{11\cdot2})$$

$$\underline{Y_2} \sim N(\qquad)$$

Conditional $dist^n$

$X_1 \mid X_2$ then the transformation.

15/01/24

Conditional distribution

(If there are two r.v.s. which are ind. then how do we define conditional distribution)

$$f(Y_1|Y_2) = \frac{f(Y_1, Y_2)}{f(Y_2)} = \frac{f(Y_1).f(Y_2)}{f(Y_2)} = f(Y_1)$$

$$\underline{X} \sim N_p(\underline{\mu}, \underline{\Sigma})$$

$$\underline{X} = \begin{pmatrix} \underline{X_1} \\ \underline{X_2} \end{pmatrix} \begin{matrix} q\times 1 \\ (p-q)\times 1 \end{matrix}$$

$\underline{Y_1} = \underline{X_1} + B\,\underline{X_2}$, (choose B such that Y_1 & Y_2 are uncorrelated)

$\underline{Y_2} = X_2$,

$$B = -\Sigma_{12}\Sigma_{22}^{-1}$$

$f(Y_1, Y_2) = f(X_1, X_2)\ |J| \qquad [|J| = 1]$

$$f(Y_1, Y_2) = N_{Y_1}(\underline{y_1},\ \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}.\mu_2,\ \Sigma_{11.2}).\ N_{Y_2}(\underline{y_2}, \mu_2, \Sigma_{22})$$

$$f(\underline{X_1}|\underline{X_2}) = \frac{f(\underline{X_1}, \underline{X_2})}{f(\underline{X_2})} = \frac{f(Y_1, Y_2)}{f(Y_2)}$$

$f_{\underline{X_1}, \underline{X_2}}(\underline{x_1}, \underline{x_2}) = f(X_1, X_2)$

$$f(X_1|X_2) \sim N_q(\mu_1 + (\Sigma_{12}\Sigma_{22})^{-1}(\underline{x_2} - \underline{\mu_2}),\ \Sigma_{11.2})$$

$f(\underline{X_1}.\underline{X_2}) = f(Y_1.Y_2) = f(Y_1)f(Y_2)$

$f(\underline{X_2}) = f(Y_2)$

$$f(X_1, X_2) = f(Y_1, Y_2)\ |J| = f(Y_1).f(Y_2)$$

$$f(X_1/X_2) = \frac{f(X_1, X_2)}{f(X_2)} = \frac{f(Y_1, Y_2)}{f(Y_2)}$$

$Y \qquad (X_1, \ldots, X_p)$

$(Y_1, \ldots, Y_q) \qquad (X_1, \ldots, X_p)$

corr

corr. coeff b/w two set of variables then it is called multiple corr. coeff.

Now $q = 1$, Y $(X_1, \ldots, X_p)$

partial coeff (corr. coeff.), we fix one variable

$r \to$ 'ρ' $\qquad \hat{\rho} = r$

$$\Sigma (x_i - \bar{x})(y_i - \bar{y}) \Big/ \sqrt{\Sigma (x_i - \bar{x})^2 \; \Sigma (y_i - \bar{y})^2}$$

one parameter can be estimated by more than one parameter

$$\underset{n \times 1}{Y_1} = X_1 + B X_2$$

assuming Y is random,
X is random so Y is also random
how does it look like ?
linear regression.

$$\begin{pmatrix} Y_1 = \underline{X_1}\,\underline{\beta_1} + \underline{\Sigma_1} \\ Y_2 = \underline{X_2}\,\underline{\beta_2} + \underline{\Sigma_2} \\ \vdots \\ Y_q = \underline{X_q}\,\underline{\beta_q} + \Sigma_q \end{pmatrix}$$

p-var.

1 variable

regression coeff.

$$\underset{\sim}{Y} \sim N(X\beta, \sigma^2 I)$$

$$Y = X\beta + \Sigma$$

$$\hat{\beta} = (\qquad)$$

$$(\underline{Y}) = X B + \varepsilon$$

$\hookrightarrow$ is a matrix of $(\beta_1, \ldots, \beta_p)$

different multiple linear regression model / multivariate regressor model

$$\hat{\beta} = (\hat{\beta}_1, \ldots, \hat{\beta}_p)$$

$$Y_1 \quad Y_2 = X_1 \beta_1 + \ldots + X_p \beta_p + \varepsilon \qquad \text{UP}$$

$$Y_1 \quad Y_2 = X_1 \beta_1 + \ldots + X_p \beta_p + \varepsilon \qquad \text{MP}$$

$$\underset{(n_1+n_2) \times 1}{\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}} = \underset{(n_1+n_2) \times p}{X}$$

B all individual corr. coeff. combined at one place.

$$Y \sim N_p(\underline{\mu}, \Sigma)$$

one — $\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N_p \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{p2} \\ \Sigma_{2p} & \Sigma_{22} \end{pmatrix} \right)$

scalar

row vector

matrices

$$\sum_{\alpha=1}^{N} (x_\alpha - \mu)' \Sigma^{-1} (x_\alpha - \mu)$$
$$= tr\left[\sum_{\alpha=1}^{N} (x_\alpha - \mu) \, \Sigma^{-1} (x_\alpha - \mu)'\right]$$
$$= tr\left[\sum_{\alpha=1}^{N} \Sigma^{-1} (x_\alpha - \mu)(x_\alpha - \mu)'\right]$$
$$= tr\, \Sigma^{-1}\left[\sum_{\alpha=1}^{N} (x_\alpha - \mu)(x_\alpha - \mu)'\right]$$
$$= tr\, \Sigma^{-1}\left(A + N(\bar{x} - \mu)(\bar{x} - \mu)'\right)$$
$$= tr\, \Sigma^{-1} A + N(\bar{x} - \mu)' \Sigma^{-1} (\bar{x} - \mu)$$ Now into scalar

Lemma 3

If D is positive definite of order $p \times p$, the ~~matrix~~ maximum of
$$f(G) = -N.\log|G| - tr\, G^{-1} D$$
w.r.t. a p.d. matrix G exists and occurs at $G = \frac{1}{N}.D$

4) Cholskey decomposition

If A is p.d., $\exists$ a unique lower triangular matrix T ($t_{ij} = 0$, $i < j$) with positive diagonal elements s.t. $A = TT'$.

~~eg~~.
$$T = \begin{pmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{pmatrix}$$

$$|T|^2 = \begin{vmatrix} a^2 & 0 & 0 \\ ab + bd & d^2 & 0 \\ ac + be + cf & de + ef & f^2 \end{vmatrix} = a^2 d^2 f^2$$

$$\log |T|^2 = \log a^2 + \log d^2 + \log f^2$$

$$TT' = \begin{bmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{bmatrix}\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix} = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 + d^2 & bc + de \\ ac & bc + de & c^2 + e^2 + f^2 \end{bmatrix}$$

$$tr\, TT' = a^2 + b^2 + c^2 + d^2 + e^2 + f^2.$$

<u>Proof.</u> Let $D = EE'$ $\quad (D > 0,$ positive definite$)$
and $E'G^{-1}E = H,$ then $G = \_\_\_\_.$

$$E'G^{-1}E = H$$

$$(E'G^{-1}E)^{-1} = H^{-1} \quad \Rightarrow \quad E^{-1}GE' = H^{-1}$$

$$\Rightarrow EE^{-1}GE'E'^{-1} = EH^{-1}E'^{-1} \quad \Rightarrow \quad G = EH^{-1}E'$$

$$|G| = |E|\,|H^{-1}|\,|E'|$$

$$= |H^{-1}|\,|EE'| \quad = \frac{|D|}{|H|}$$

$$\operatorname{tr}(G^{-1}D) = \operatorname{tr}(G^{-1}EE') \quad = \operatorname{tr}(E'G^{-1}E) \quad = \operatorname{tr}(H)$$

$$f(G) = -N\log|G| - \operatorname{tr} G^{-1}D$$

$$= -N\log\frac{|D|}{|H|} - \operatorname{tr} H$$

H is lower triangular

$$= -N\log|D| + \underline{N\log|H| - \operatorname{tr} H}$$

using Cholskey dec.

$(H = TT')$

$$= -N\log|D| + N\log|T|^2 - \operatorname{tr} TT'$$

$$= -N\log|D| + \sum_{i=1}^{n} N\log t_{ij}^2 - \left(\sum_{i=1}^{p} t_{ii}^2 + \sum_{i<j}\sum t_{ij}^2\right)$$

$$= -N\log|D| + \sum_{i=1}^{n}\left(N(\log t_{ii}^2) - t_{ii}^2\right) - \sum_{i<j}\sum t_{ij}^2$$

differentiate matrix w.r.t. scalars

$$\frac{\partial f}{\partial t^2} = \frac{N}{t_{ii}^2} - 1 = 0 \quad \Rightarrow N = t_{ii}^2$$

$$\frac{\partial^2 f}{\partial t_{ii}^2} = -\frac{N}{t_{ii}^2} \quad = -\frac{1}{t_{ii}^2} < 0$$

$$E(s^2) = \sigma^2$$

$$E(s_x^2) = \sigma_x^2 \quad \text{and} \quad E(s_y^2) = \sigma_y^2$$

$$S = \begin{pmatrix} s_x^2 & s_{xy} \\ s_{xy} & s_y^2 \end{pmatrix} \qquad E(S) = ?$$

$$E(S) = E\begin{pmatrix} s_x^2 & s_{xy} \\ s_{xy} & s_y^2 \end{pmatrix} = \begin{pmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{pmatrix} = \Sigma$$

$N = t_{ii}^2$

$H = NI$

$$\begin{pmatrix} t_1^2 & & \\ & t_2^2 & \\ & & \ddots \end{pmatrix}$$

$$N = (n_{ij}) \cdot \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

Then $G = E H^{-1} E'$

$$= \frac{1}{N} EE' = \frac{1}{N} \cdot D$$

$$f(G) = -N \log |G| - tr\, G^{-1} D$$

$$f\left(\frac{1}{N} \cdot D\right) = -N \log \left|\frac{D}{N}\right| - tr\, D^{-1} D N$$

$$= -N \log \left|\frac{D}{N}\right| - N\, tr\, D^{-1} D$$

$tr\, D^{-1} D = tr\, I_p$

$= p$

$$= -N \log |D| - pN \log N - Np$$

after removing $|N| = N^p$

$$\hat{A} = \begin{pmatrix} \Sigma (x_i - \hat{\mu})^2 & 0 \\ 0 & \Sigma (y_i - \hat{\mu})^2 \end{pmatrix}$$

max. value of this fn.

(will find out using elements)

$t_{ii}^2 = N_{ii}$

$$\begin{pmatrix} N_{11} & & & \\ & N_{22} & & \\ & & \ddots & \\ & & & N_{pp} \end{pmatrix} = \begin{pmatrix} N & & \\ & \ddots & \\ & & N \end{pmatrix} = N(I)$$

$E(x'Ax) = \sigma^2 tr\, A$

$= E(x'Ax \cdot x'Bx \cdot xx')$

$$\begin{pmatrix} a_{11} & a_{12} & & \\ & & \ddots & \\ & & & a_{pp} \end{pmatrix}$$

$$\begin{pmatrix} a_{11}b_{11} & & & \\ & \ddots & & \\ & & a_{ij}b_{ij} & \\ & & \ddots & \\ & & & a_{pp}b_{pp} \end{pmatrix}$$

$$A = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \qquad B = \begin{pmatrix} b_1 & b_2 \\ b_3 & b_4 \end{pmatrix}$$

$$A * B = \begin{pmatrix} a_1 b_1 & a_2 b_2 \\ a_3 b_3 & a_4 b_4 \end{pmatrix}$$

$$A \circledast B = \begin{pmatrix} a_1 B & a_2 B \\ a_3 B & a_4 B \end{pmatrix}$$

Maximum likelihood Estimation (MLE)

$$\underset{\sim}{x_1}, \dots, x_N \sim N_p(\underline{\mu}, \Sigma) \qquad N > p$$

$$L = \frac{1}{(2\pi)^{Np/2} \, |\Sigma|^{N/2}} \exp\left[-\frac{1}{2}\sum_{\alpha=1}^{N} (\underline{x_\alpha} - \mu)' \Sigma^{-1} (x_\alpha - \mu)\right]$$

$$\log L = \underbrace{-\frac{Np}{2} \log 2\pi}_{\text{constant}} - \frac{N}{2} \log|\Sigma| - \underbrace{\frac{1}{2}\sum_{\alpha=1}^{N} (x_\alpha - \mu)' \Sigma^{-1} (x_\alpha - \mu)}_{\text{Result } \underline{2}}$$

$$= \underline{-\frac{Np}{2} \log 2\pi} - \underbrace{\frac{N}{2} \log|\Sigma|}_{\text{I}} - \underbrace{\frac{1}{2} \operatorname{tr} \Sigma^{-1} A}_{\text{II}}$$

$$- \underbrace{\frac{N}{2} (\bar{x} - \mu)' \Sigma^{-1} (\bar{x} - \mu)}_{\text{III}}$$

III) Minimize it because it has -ve sign

$\Sigma > 0$, So $\Sigma^{-1} > 0$,

$$(\bar{x} - \mu)' \Sigma^{-1} (\bar{x} - \mu) \geq 0$$

$$= 0 \quad \text{iff} \quad \mu = \bar{x}$$

Lemma

$$f(\Sigma) = -N \log|\Sigma| - \operatorname{tr} \Sigma^{-1} A.$$

Maximum occurs at $\Sigma = \dfrac{A}{N}$

$G = \Sigma$

$D = A$

Lemma

$$f(\Sigma) = -N \log|\Sigma| - \operatorname{tr} \Sigma^{-1} A$$

Maximum occurs at $\Sigma = \frac{A}{N}$ $\qquad G = \Sigma,\ D = A$

$$\hat{\Sigma} = \frac{1}{N} \sum (x_\alpha - \bar{x})(x_\alpha - \bar{x})'$$

MLE not unbiased.

$\hat{\mu} = \bar{x}$ $\qquad S = \frac{A}{N-1}$

going to achieve the max. at $\hat{\mu} = \bar{x}$.

$$E(\hat{\mu}) = E(\bar{x}) = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} = \underline{\mu}$$

$$S = \begin{pmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ \vdots & \ddots & & \\ & & \ddots & \\ s_{p1} & s_{p2} & \cdots & s_{pp} \end{pmatrix}$$

$$E(s_x^2) = \sigma_x^2$$

$$E(s_{xy}) = \sigma_{xy}$$

$$E(S) = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & & \\ & \sigma_2^2 & & \sigma_{ij} \\ & & \ddots & \\ \sigma_{ji} & & & \sigma_p^2 \end{pmatrix} = \Sigma$$

17/01/24

Multivariate distⁿ in R

$\underset{20}{X}$ $p = 3 \sim N(\mu, \Sigma)$ $\qquad \bar{x},\ \hat{\Sigma}$

$\hookrightarrow$ change p, n

$\bar{x} - \underline{\mu}$

change the structure of cov-matrix (Σ)

if Σ is very high and n is small then result is very bad

From the simulation exp., you get the feeling of the data

17/01/24

Distribution of sample mean vector

$$\underline{X} \sim N_p(\underline{\mu}, \Sigma)$$

$$(\underline{X}_1, \ldots, \underline{X}_N) = \frac{1}{(2\pi)^{\frac{Np}{2}} \; |\Sigma|^{N/2}} \exp\left[-\frac{1}{2}\sum_{i=1}^{N}(\underline{x}_\alpha - \mu)' \Sigma^{-1} (\underline{x}_\alpha - \mu)\right]$$

Apply the transformation,

$X^* = XQ$, Q is the orthogonal matrix s.t. $(QQ' = I)$ the first column has element $\frac{1}{\sqrt{N}}$ and all other are q_{ij}.

$$Q = \left(\begin{array}{c|c} \frac{1}{\sqrt{N}} & \\ \vdots & q_{ij} \\ \frac{1}{\sqrt{N}} & \end{array}\right) \qquad X = \begin{pmatrix} X_{11} & \cdots & X_{1N} \\ \vdots & \ddots & \\ X_{p1} & & X_{pN} \end{pmatrix}$$

$$X^* = XQ$$

$= [\sqrt{N}\, \overline{\underline{X}}, \; Y]$ → Reduced matrix (remaining part of the matrix)

$$X^* X^{*\prime} = XQ.Q'X' = XX'$$

$$X^* X^{*\prime} = N \overline{\underline{X}}\, \overline{\underline{X}}' + YY' \longrightarrow [\sqrt{N}\,\overline{\underline{X}}, Y][\sqrt{N}\,\overline{\underline{X}}, Y]' = N\overline{X}\,\overline{X}' + YY'$$

$$YY' = \underline{X}\,\underline{X}' - N\overline{\underline{X}}\,\overline{\underline{X}}'$$

$$\sum_{\alpha=1}^{N} \underline{y}_\alpha \underline{y}_\alpha' = \sum_{\alpha=1}^{N}(\underline{x}_\alpha - \overline{\underline{x}})(\underline{x}_\alpha - \overline{\underline{x}})'$$

$$= A \quad = (N-1)S$$

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}(x_i^2 + \bar{x}^2 - 2x_i\bar{x})$$

$$= \sum_{i=1}^{n} x_i^2 + n\bar{x}^2 - 2\bar{x}\, n\bar{x}$$

$$= \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

$$(\underline{X}_1, \ldots \underline{X}_N) = \underbrace{\frac{1}{(2\pi)^{\frac{Np}{2}} |\Sigma|^{N/2}}}_{K^*} \exp\left[-\frac{1}{2}\sum_{i=1}^{N} (\underline{x}_\alpha - \underset{\pm \underline{\bar{x}}}{\underline{\mu}})' \, \Sigma^{-1} (\underline{x}_\alpha - \underline{\mu})\right]$$

$$= K^* \exp\left[-\frac{1}{2} \operatorname{tr} \Sigma^{-1} \sum_{\alpha=1}^{N} \underbrace{(\underline{x}_\alpha - \underline{\mu})(\underline{x}_\alpha - \underline{\mu})'}_{\pm \bar{x}}\right]$$

$$= K^* \exp\left[-\frac{1}{2} \operatorname{tr} \Sigma^{-1} \left\{ \sum_{\alpha=1}^{N} (x_\alpha - \bar{\underline{x}})(x_\alpha - \bar{x})' + N(\bar{\underline{x}} - \underline{\mu})(\bar{\underline{x}} - \underline{\mu})' \right\}\right]$$

$$= K^* \exp\left[-\frac{1}{2} \operatorname{tr} \Sigma^{-1} \{ YY' + N(\bar{\underline{x}} - \underline{\mu})(\bar{\underline{x}} - \underline{\mu})' \}\right]$$

$$= \frac{1}{(2\pi)^{\frac{Np}{2}} |\Sigma|^{N/2}} \exp\left[-\frac{1}{2} \operatorname{tr} \Sigma^{-1} YY\right] \times \quad \to \text{scalar} \quad Y'\Sigma^{-1}Y$$

$$\exp\left[-\frac{1}{2} (\underline{x} - \underline{\mu})' \left(\frac{\Sigma}{N}\right)^{-1} (\underline{x} - \mu)\right]$$

$$= \frac{1}{(2\pi)^{\frac{Np}{2}} |\Sigma|^{N/2}} \exp\left[-\frac{1}{2} Y'\Sigma^{-1}Y\right]$$

$f(Y)$

$$\exp\left[-\frac{1}{2} (\underline{x} - \underline{\mu})' \left(\frac{\Sigma}{N}\right)^{-1} (\underline{x} - \mu)\right]$$

$f(x)$

$$\sim N_p\left(\mu, \frac{\Sigma}{N}\right)$$

we can say X and Y are independent.

1) $\underline{\bar{X}}$ and Y are independently distributed.

2) $\underline{\bar{X}} \sim N_p\left(\underline{\mu}, \frac{\Sigma}{N}\right)$

3) (Y is a linear transformation of X)

$\underline{Y} \sim N(0, \Sigma)$ $\qquad\qquad\qquad\qquad\qquad\qquad \underline{Y} \sim N(\quad)$

$Y'\Sigma^{-1}Y \longrightarrow (Y-0)'\Sigma^{-1}(Y-0)$

vector

Y : matrix $\longrightarrow$ wishart dist^n.

$Y : Y = (N-1)S$

$$f(A) = \frac{|A|^{\frac{N-p-1}{2}} \exp\left[-\frac{1}{2} \operatorname{tr} \Sigma^{-1} A\right]}{2^{\frac{Np}{2}} \, \pi^{\frac{p(n-1)}{2}} \, |\Sigma|^{N/2} \prod_{i=1}^{n-1} \Gamma\left(\frac{N-i}{2}\right)}$$

Γ → Gamma fn.

$Q_1 = \underline{X}' A \underline{X}$

$Q_2 = \underline{X}' B \underline{X}$

$Q_3 = \underline{X}' C \underline{X}$

$$\begin{pmatrix} | & | & | \end{pmatrix}$$

$Q = X'AX$

column vector will form a matrix

$\underline{X} \sim N_p(\mu, \Sigma)$

$\underline{X}'\Sigma^{-1}\underline{X} \sim$ Non central $\chi^2(p, \mu'\Sigma^{-1}\mu)$

C: non-singular matrix s.t. $C\Sigma C' = I$.

$Z = C\underline{X} \sim N(C\underline{\mu}, C\Sigma C')$

$\sim N(\underline{\lambda}, I)$

$X'\Sigma^{-1}\underline{X} = Z'C'^{-1}\Sigma^{-1}C^{-1}Z$

$= Z'(C\Sigma C')^{-1} Z \quad = Z'Z$

Distribution of $X'\Sigma^{-1}X \sim \sum_{i=1} Z_i Z \sim NC\ \chi^2(p, \Sigma\lambda_i^2)$

$$(\bar{X}-\mu)\left(\frac{\Sigma}{N}\right)^{-1}(X'-\mu) \sim \chi^2_{(p)}$$

$$\sqrt{N}(\bar{X}-\mu) \sim N(0, \Sigma)$$

$$\boxed{N(\bar{X}-\mu)'\Sigma^{-1}(\underline{\bar{X}}-\mu) \sim \chi^2_{(p)}}$$

$n(x-\bar{x})\, s^2 (x-\bar{x})$

$$\frac{(x-\bar{x})^2}{\frac{s^2}{n}}$$

When Σ is known, $H_0 : \underline{\mu} = \underline{\mu_0}$

(then it is used to test)

When Σ is unknown, Hotelling T^2.

23/01/24

Wishart distribution :

$x_1, \ldots, x_n \sim N(\mu, \Sigma)$, $\hat{\mu} = \bar{x}$, $\hat{\Sigma} = S = \dfrac{A}{N-1} = \dfrac{A}{n}$

$$A = \sum_{\alpha=1}^{N} (\underline{X_\alpha} - \bar{\underline{X}})(\underline{X_\alpha} - \bar{\underline{X}})',$$

positive definite

$n = N-1$,
$(N-1)S = A$

$$S = \begin{pmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ \vdots & s_{22} & & \vdots \\ \vdots & & & \vdots \\ s_{p1} & s_{p2} & \cdots & s_{pp} \end{pmatrix}$$

$\dfrac{p(p+1)}{2}$ elements
have unique elements

Let $X_1, \ldots, X_N$ $(N > p)$ are ind of each other, $N(\underline{\mu}, \Sigma)$.
If a random positive definite matrix A has the pdf

$$f(A) = \frac{|A|^{\frac{n-p-1}{2}} \exp\left[-\frac{1}{2} \operatorname{tr} \Sigma^{-1} A\right]}{2^{\frac{np}{2}} |\Sigma|^{n/2} \pi^{\frac{p(p-1)}{4}} \prod_{i=0}^{p-1} \Gamma\left(\frac{n-i}{2}\right)}$$

then A is said to have a wishart distribution with parameter Σ and n.
$A \sim W(\Sigma, n)$.

Theorem If $\underline{y_\alpha} \sim N_p(0, \Sigma)$ $\alpha = 1, \ldots, n$ and are ind. (independently) distributed then the density of

$$A = \sum_{\alpha=1}^{n} \underline{y_\alpha}\, \underline{y_\alpha'} \sim W(\Sigma, n)$$

for positive definite A and $n > p$. No density if $n < p$.
(assuming that A is a p.d. otherwise we can use generalised inverse)

Theorem If A and B are two real symmetric matrices s.t. B is p.d., then there exists a n.s matrix P s.t. $P'AP = \text{diag}(\lambda_1, \ldots, \lambda_n)$, $P'BP = I_n$ where $\lambda_1, \ldots, \lambda_n$ are the roots of the eqn $|A - \lambda B| = 0$

Proof B is p.d., $\exists$ a n.s. matrix Q s.t. $Q'BQ = I$.
Consider $Q'AQ$: real, symm, so by spectral decomposition thm, $\exists$ an orthogonal matrix H s.t. $H'(Q'AQ)H = D$

$$H'(A'QA)H = D = \text{diag}(\lambda_1, \ldots, \lambda_n)$$ where $\lambda_1, \ldots, \lambda_n$ are the roots of $|Q'AQ - \lambda I| = 0$

). Let $P = QH$ (P is n.s.) $\Rightarrow P'AP = D = \text{diag}(\lambda_1, \ldots, \lambda_n)$
Q : ns $\Rightarrow P$ is also ns.
$Q'BQ = I$ or $H'QBQH = H'H$ or $P'BP = I$.

Let $P = QH$, (P is n-s)

$$\Rightarrow P'AP = D = \text{diag}(\lambda_1, \ldots, \lambda_n)$$

Since Q is n-s, so p is also n-s.
Since $Q'BQ = I$

$$H'Q'BQH = H'H$$

$$\Rightarrow P'BP = I_n$$

Combine $|Q'AA - \lambda I| = 0$

$$\sim |Q'AQ - \lambda Q'BQ| = 0$$

$$\sim |Q'|\,|A - \lambda B|\,|Q| = 0$$

$$\sim |QQ'|\,|A - \lambda B| = 0$$

$$|A - \lambda B| = 0$$

$y \sim N(0, \Sigma)$ $\quad$ $(\ \ ) \sim W(\Sigma, n$

Characteristic function

$$\underline{y}_\alpha \overset{ind}{\sim} N(0, \Sigma), \quad \alpha = 1, \dots, n, \quad A = \sum_{\alpha=1}^{n} \underline{y}_\alpha \underline{y}_\alpha'$$

Ch. fn. $\phi(\Theta) = E[\exp(i \operatorname{tr} A \Theta)]$

$$= E\left[\exp\left(i \operatorname{tr} \sum_{\alpha=1}^{n} \underline{y}_\alpha \underline{y}_\alpha' \Theta\right)\right]$$ — (scalar) because of trace

$$= E\left[\exp\left(i \sum_{\alpha=1}^{n} \underline{y}_\alpha' \Theta \underline{y}_\alpha\right)\right]$$

$$= \prod_{\alpha=1}^{n} E\left[\exp\left(i \underline{y}_\alpha' \Theta \underline{y}_\alpha\right)\right]$$

$B = \Sigma$

$A = \Theta$

$$= \left\{E\left[\exp\left(i \underline{y}' \Theta \underline{y}\right)\right]\right\}^n$$

Consider $E[\exp(i\underline{y}'\Theta\underline{y})]$

consider Σ^{-1} and for all Θ real, $\exists$ a real n.s. matrix P s.t. $P'\Sigma^{-1}P = I$, $P'\Theta P = \operatorname{diag}(\lambda_1, \dots, \lambda_p) = D$ where $\lambda_1, \dots, \lambda_p$ are the ch. roots of $|\Theta - \lambda I| = 0$

Now $\underline{z} \sim N(0, I)$

Consider a linear transformation $\underline{y} = P\underline{z}$, z_j are ind-dist.

$$E[\exp(i\underline{y}'\Theta\underline{y})] = E[\exp(i\underline{z}'\underbrace{P'\Theta P}_{D}\underline{z})]$$

$$= E\left[\exp\left(i \sum_{j=1}^{p} \lambda_j z_j^2\right)\right]$$

$z_j \sim N(0,1)$

order $= 1$

$z_j^2 \sim \chi^2_{(1)}$

[ch. of $\chi^2_{(1)}$] $$= \prod_{j=1}^{p} (1 - 2i\lambda_j)^{-1/2}$$

$$= |I - 2iD|^{-1/2}$$

$$= |P'\Sigma^{-1}P - 2i P'\Theta P|^{-1/2}$$

$$= [|P'|\,|\Sigma^{-1} - 2i\Theta|\,|P|]^{-1/2}$$

$$= \frac{|\Sigma^{-1} - 2i\Theta|^{-1/2}}{|P'P|^{1/2}} = \frac{|\Sigma^{-1} - 2i\Theta|^{-1/2}}{|\Sigma|}$$

$$= |I - 2i\Theta\Sigma|^{-1/2}$$

$\phi(\Theta) = |I - 2i\Theta\Sigma|^{-n/2}$ ~ fn. of Σ, n.

$f(A) =$ ______

$\phi_A(\Theta) = \int f(A) \exp(i \, tr \, A\Theta) \, dA$

Additive Property

If $A_j \ (j=1,\dots,q) \overset{ind}{\sim} W(\Sigma, n_j)$

then $A = \sum_{j=1}^{q} A_j \sim W\left(\Sigma, \sum_{j=1}^{q} n_j\right)$

$X \sim N(0, \Sigma_1^2)$

$Y \sim N(0, \Sigma_2^2)$

$X^2 + Y^2 \sim \chi^2_{(\quad)}$

$\phi_{A_j}(\Theta) = |I - 2i\Theta\Sigma|^{-\frac{n_j}{2}}$

$\phi_A(\Theta) = \phi_{\Sigma A_j}(\Theta) = \Pi \, \phi_{A_j}(\Theta)$

$= |I - 2i\Theta\Sigma|^{-\sum_{j=1}^{q}\frac{n_j}{2}}$

$= |I - 2i\Theta\Sigma|^{-\frac{n_1}{2}} \times \dots \dots \times |I - 2i\Theta\Sigma|^{-\frac{n_q}{2}}$

$= |I - 2i\Theta\Sigma_1|^{-\frac{n_1}{2}} \times \dots \dots \times |I - 2i\Theta\Sigma_q|^{-\frac{n_q}{2}}$

$A \sim W\left(\Sigma, \sum_{j=1}^{q} n_j\right)$ (Reproductive property)

(Fisher behaviour problem in a multivariate case while testing of Hypo.)